

⚡ BNI Enterprises - Bike Dealer Management System - Comprehensive Feature Report

📁 Module & Features	🌟 Key Capabilities	🔒 Security & Access
🏠 Dashboard & Analytics ✓ Role-specific statistics cards ✓ Live metrics: Stock, Sold, Returned, Damaged, Purchase/Sales Value, Tax, Profit ✓ Interactive charts: Sales Trend, Model-wise Stock, Income vs Expense, Inventory Status ✓ Quick action shortcuts for core workflows ✓ Alert banners for pending cheques & overdue installments ✓ Recent sales/purchases timeline	📊 Real-time visual insights with Chart.js 📱 Fully responsive mobile-first layout ⚡ Instant data loading with cached queries 🎨 Animate.css micro-interactions for engagement	🔒 Role-based dashboard visibility 👁️ Data filtered by user permissions 🔄 Auto-session refresh prevents timeout
🔑 Authentication & Session Management ✓ Secure login with math-based SVG CAPTCHA ✓ Password hashing (bcrypt via password_hash) ✓ Session regeneration on successful login ✓ IP-based ban system (7 failed attempts = 3hr lockout) ✓ Configurable idle (40min) & absolute (8hr) timeouts ✓ CSRF token validation on all POST requests	📱 Dynamic CAPTCHA with +/– operators 🔄 Auto-CAPTCHA refresh on click 🕒 Graceful session expiry with redirect 🛡️ Multi-layer request validation	🔒 session.cookie_httponly + SameSite=Strict 🚫 Rate limiting via file-based IP tracking 🔑 Strong password policy enforcement (8+ chars, special, number, case)
📦 Purchase Entry Module ✓ Multi-bike purchase orders with supplier linkage ✓ Chassis/Motor number tracking with uniqueness validation ✓ Model, color, purchase price, safeguard notes per bike ✓ Image upload with auto-resize & compression (max 400KB) ✓ Integrated payment tracking (cash/cheque/bank/online)	🖼️ Image preview with lightbox support 🔍 AJAX chassis uniqueness check before save ✚ Modal forms to add suppliers/models without leaving page 📊 Real-time payment vs purchase price difference display	🔒 Prepared statements prevent SQL injection ✓ File upload validation: extension, size, image verification 📄 Transaction rollback on partial save failure

 Module & Features

- ✓ Cheque details: number, bank, date, status workflow
- ✓ Auto-divide total payment across bikes toggle
- ✓ Tax calculation based on configurable base (purchase/selling price)

 Key Capabilities **Security & Access** **Inventory/Stock Management**

- ✓ Comprehensive bike listing with advanced filtering
- ✓ Status workflow: in_stock → sold → returned → damaged_lost
- ✓ Visual status badges with color-coded rows
- ✓ Edit bike details: color, price, status, notes, image
- ✓ Bulk delete with confirmation & status validation
- ✓ CSV export with all applied filters
- ✓ Individual bike view with complete history timeline
- ✓ Page-level totals: Purchase Price, Selling Price, Margin

-  Multi-field search: chassis, motor, model, color
-  Date-range filtering for inventory date
-  Print-optimized view with hidden UI elements
-  "Days in Stock" calculation for aging analysis

-  Delete protection: cannot remove sold/returned bikes
- ✓ Status transition validation prevents invalid workflows
-  Audit-ready data display with full transaction history

 Sales Entry Module

- ✓ In-stock bike selection with auto-fill purchase price
- ✓ Dynamic margin calculation: Selling – Purchase – Tax
- ✓ Customer selection with Filer/Non-Filer status display
- ✓ Walk-in customer support with full-payment enforcement
- ✓ Down payment with multi-method support (cash/cheque/bank/online)
- ✓ Installment plan creation: total count, amount, first due date
- ✓ Accessories bundling: quantity, unit price, discount, final price
- ✓ Auto-stock deduction for sold accessories
- ✓ Professional invoice generation with QR-ready layout

-  Real-time tax/margin/remaining balance calculations
-  Custom accessory entry creates new record if not found
-  Mobile-optimized form with collapsible sections
-  Thermal/letterhead compatible invoice printing

-  Walk-in restriction: partial payments blocked
- ✓ Accessory stock validation prevents overselling
-  Transaction atomicity: sale + payment + ledger + installments in one commit

**Module & Features****Key Capabilities****Security & Access****↩ Returns Management**

- ✓ Dual-tab interface: Sales Returns & Purchase Returns
- ✓ Sales Return: refund amount, method, notes, full reversal logic
- ✓ Purchase Return: supplier linkage, refund received tracking
- ✓ Cheque refund support with full payment workflow
- ✓ Automatic status update: sold → returned / in_stock → returned_to_supplier
- ✓ Accessory price inclusion in total reversal calculation



One-click return processing with validation



Full vs partial refund handling with ledger reconciliation



Return notes for audit trail & customer communication



Status validation: only sold bikes can be returned by customers



Reference integrity: prevents return of already-returned items



Dual ledger entries for reversal + refund tracking

**Payments Register**

- ✓ Unified payment listing across all transaction types
- ✓ Filter by: cheque status, transaction type, bank, date range
- ✓ Cheque workflow: pending → cleared/bounced/cancelled
- ✓ Bounced cheque auto-reversal: ledger + installment adjustments
- ✓ Summary cards: pending/cleared/bounced counts & amounts
- ✓ Payment deletion with confirmation & referential checks



Visual status badges with color coding



Advanced filtering for reconciliation workflows



Party name display for quick identification



Bounced cheque processing requires admin permission



Installment recalculation on payment reversal



Complete payment audit trail with reference linking

**Installments Management**

- ✓ Installment listing with customer, bike, due date, status
- ✓ Status workflow: pending → paid / overdue → cancelled
- ✓ Payment modal: amount paid, penalty fee, payment method
- ✓ Cheque details support for installment payments
- ✓ Auto-status update when amount_paid ≥ installment_amount
- ✓ Customer-wise filtering for account management



Penalty fee addition to total payment amount



Due date highlighting for overdue identification



Remaining balance display per installment



Payment recording requires edit permission



Amount validation: cannot exceed remaining due



Payment linkage to ledger for double-entry accounting

**Quotations Module**

- ✓ Quotation creation: customer,



Seamless quotation-to-sale workflow



Conversion requires both quotations & sales permissions

 Module & Features

bike, accessories, price, validity

- ✓ Accessory bundling with quantity/discount/final price
- ✓ One-click conversion to sale: creates sale entry + updates inventory
- ✓ Status tracking: pending → accepted/rejected/converted
- ✓ Edit/delete with permission checks
- ✓ Customer & bike detail preview in quotation form

 Key Capabilities

-  Accessory pricing auto-fill from inventory
-  Validity date enforcement for quote expiration

 Security & Access

- ✓ Stock validation before conversion (bike must be in_stock)
-  Status change logging for audit compliance

 Customer Management

- ✓ CRUD operations: name, phone, CNIC, filer status, address
- ✓ Pakistan-specific CNIC format display & validation hint
- ✓ Filer/Non-Filer badge for tax compliance tracking
- ✓ Purchase history: bike count & total amount spent
- ✓ Search by name, phone, or CNIC
- ✓ Delete protection if bikes are linked

-  Click-to-call phone number display
-  CNIC monospace font for readability
-  Direct link to customer ledger for account history

-  Delete validation: prevents removal of customers with sales
- ✓ Input sanitization prevents XSS in display fields
-  Audit trail via ledger for all customer transactions

 Supplier Management

- ✓ CRUD operations: name, contact, address
- ✓ Purchase order count & total value display
- ✓ Direct link to supplier ledger for payment history
- ✓ Delete protection if purchase orders exist
- ✓ Modal quick-add from purchase entry page

-  Supplier performance metrics at a glance
-  Seamless navigation to related transactions
-  Contextual add-from-purchase workflow

-  Referential integrity: blocks deletion with active orders
- ✓ Prepared statements prevent injection in search/sort
-  Purchase order linkage for complete audit trail

 Model Management

- ✓ CRUD: model code, name, category, short code, image
- ✓ Inventory metrics: total bikes, in-stock count, sold count
- ✓ Quick-action buttons: purchase/sell pre-filtered by model
- ✓ Image upload with preview & compression

-  Model image gallery for visual identification
-  Stock vs sales ratio display per model
-  Pre-filtered purchase/sale forms for workflow efficiency

-  Delete validation: prevents removal of models with inventory
- ✓ Image upload: extension, size, MIME type validation
-  Model usage tracking for business intelligence

 Module & Features **Key Capabilities** **Security & Access**

- ✓ Delete protection if bikes are linked

 Accessories Management

- ✓ CRUD: name, SKU, purchase price, selling price, stock
- ✓ Stock valuation: total purchase value & selling value
- ✓ Sales analytics: quantity sold, revenue, profit, discounts given
- ✓ Top-sold accessories bar chart visualization
- ✓ Delete protection if sales records exist

-  Real-time profit margin calculation per accessory
-  Visual ranking of best-selling items
-  Estimated future profit based on current stock

-  Stock deduction validation prevents negative inventory
- ✓ Price validation: non-negative values enforced
-  Sales linkage tracking for complete product lifecycle

 Income/Expense Module

- ✓ Entry management: date, type, category, amount, method, notes
- ✓ Dynamic category suggestions from existing entries
- ✓ Filtering: type, category, date range
- ✓ Summary cards: total income, expense, net balance, avg transaction
- ✓ Visual charts: category breakdown, daily trend comparison
- ✓ Created-by tracking for accountability

-  Category-based color coding for quick scanning
-  Date-range trend analysis for financial planning
-  Category autocomplete for consistent data entry

-  Edit/delete permissions enforced per entry
- ✓ Amount validation: positive values only
-  User attribution for all financial entries

 Customer Ledger

- ✓ Running balance calculation with debit/credit entries
- ✓ Transaction history: description, date, amount, balance
- ✓ Purchase history sub-table for customer
- ✓ Receive payment modal with auto-installment distribution
- ✓ Advance payment (loan) recording capability
- ✓ Print-optimized ledger view

-  Real-time balance with Dr/Cr indicator
-  Payment auto-allocation to oldest pending installments
-  Complete transaction narrative for dispute resolution

-  Payment recording requires customer selection validation
- ✓ Balance calculation integrity: double-entry enforcement
-  Immutable ledger entries for financial compliance

 Supplier Ledger

- ✓ Purchase order & payment tracking with running balance
- ✓ Transaction merging: orders (debit) + payments/refunds (credit)

-  Payable/Advance balance indicator with color coding
-  Direct navigation from supplier list to ledger

-  Payment recording validates supplier existence
- ✓ Balance integrity: purchase total – payment total = balance

 Module & Features

- ✓ Make payment & receive refund modals
- ✓ Print-optimized ledger for reconciliation
- ✓ Chronological sorting with date + ID tiebreaker

 Key Capabilities

-  Transaction type badges for quick identification

 Security & Access

-  Complete audit trail for supplier financials

 Reports Module

- ✓ 11 specialized sub-reports with tab navigation:
 - Current Stock: days in stock analysis
 - Sold Bikes: date-range margin reporting
 - Model-wise: inventory/sales/returns aggregation
 - Tax Report: monthly tax collection summary
 - Profit/Margin: monthly profitability with averages
 - Bank/Cheque: cheque transactions by bank & type
 - Monthly Summary: yearly purchase vs sales comparison
 - Daily Ledger: single-day comprehensive snapshot
 - Purchase vs Sales: monthly variance analysis
 - Accessory Stock: current inventory valuation
 - Installments Summary: customer-wise overdue tracking
- ✓ Date/year/month filters per report
- ✓ Print-optimized layouts for all reports

-  Interactive data aggregation with one-click filtering

-  Professional report formatting for stakeholder presentation

-  CSV export capability for external analysis

-  Drill-down from summary to transaction detail

-  Report access controlled by page-level permissions

- ✓ Data aggregation validation prevents calculation errors

-  Report generation logging for compliance auditing

 Roles & Permissions

- ✓ Role CRUD: name, description
- ✓ Granular permissions per page: view/add/edit/delete
- ✓ 17+ module permission matrices
- ✓ Administrator role protection (cannot rename/delete)
- ✓ User assignment validation before role deletion

-  Visual permission matrix with checkbox interface

-  Instant permission application without page reload

-  Role description for documentation & onboarding

-  Self-protection: cannot delete own role or admin role

- ✓ Permission validation on every protected action

-  Role change logging for security auditing

Module & Features

- ✓ Permission inheritance: Admin = all permissions

Key Capabilities

Security & Access

User Management

- ✓ User CRUD: username, password, full name, role, active status
- ✓ Password strength validation: 8+ chars, uppercase, lowercase, number, special
- ✓ Role assignment from available roles dropdown
- ✓ Active/inactive toggle for access control
- ✓ Delete protection for admin & currently logged-in user

- 🔑 Password visibility toggle for secure entry
- 📋 User list with role & status at a glance
- ⚡ Quick activation/deactivation for access management

- 🔒 Password hashing with PASSWORD_DEFAULT (bcrypt)
- ✓ Username uniqueness validation on create/update
- 📋 Session invalidation on user deactivation

System Settings

- ✓ Company configuration: name, branch, currency symbol
- ✓ Tax configuration: rate (%) + calculation base (purchase/selling)
- ✓ Invoice settings: show/hide purchase price
- ✓ Session timeout: idle & absolute duration (seconds)
- ✓ Admin password change with strength validation
- ✓ Database backup: full SQL dump download
- ✓ Database restore: SQL file upload with overwrite warning
- ✓ System info display: PHP version, MySQL version, server time

- 🌞 Live theme toggle (dark/light) with persistence
- 🕒 Timeout configuration for security compliance
- 📁 One-click backup generation for disaster recovery

- 🔒 Settings changes require edit permission
- ✓ Restore operation: multi-step confirmation + warning
- 📋 Settings change logging for configuration auditing

UI/UX Excellence

- ✓ Bootstrap-inspired responsive layout with custom CSS variables
- ✓ Dark/Light theme toggle with localStorage persistence
- ✓ Collapsible sidebar with mobile hamburger menu
- ✓ SweetAlert2 for confirmations, notifications, modals
- ✓ JustValidate for client-side form validation
- ✓ Select2 for enhanced dropdowns

- 🌞 Professional dark-theme aesthetic with accent colors
- 📱 Mobile-first responsive design with breakpoint adaptations
- ⚡ Instant visual feedback on all user actions
- 🗂️ Keyboard-navigable forms & controls
- 🖨️ Print-ready layouts for invoices, ledgers, reports

- 🔒 All forms include CSRF token hidden input
- ✓ Client-side validation mirrored by server-side checks
- 📋 User action logging via ledger & audit tables

Module & Features**Key Capabilities****Security & Access**

with search

- ✓ DataTables with pagination, sorting, responsive, export
- ✓ Chart.js for interactive data visualization
- ✓ Animate.css for subtle entry animations
- ✓ Print-optimized stylesheets hiding UI chrome
- ✓ Toast notifications for success/error feedback

Progressive Enhancements

- ✓ AJAX chassis uniqueness validation
- ✓ Dynamic supplier/model/customer dropdown population
- ✓ Real-time payment/margin/installment calculations
- ✓ Modal forms for quick entity creation without navigation
- ✓ Auto-scroll to forms after modal close
- ✓ Image preview with lightbox for bike photos

- 🔄 Seamless data updates without page reload
- ✚ Contextual quick-add workflows reduce navigation friction
- 📊 Live financial calculations prevent entry errors
- 🖼️ Visual verification of uploaded images before save

- 🔒 AJAX endpoints validate session & permissions
- ✓ Input sanitization on all dynamic content insertion
- 📝 Modal actions logged via standard form submission

🎯 Data Integrity & Business Logic

- ✓ Double-entry ledger for all financial transactions
- ✓ Referential integrity via FOREIGN KEY constraints
- ✓ Status workflow enforcement: in_stock → sold → returned
- ✓ Stock deduction on accessory sale
- ✓ Margin calculation: selling – purchase – tax
- ✓ Tax flexibility: calculated on purchase OR selling price
- ✓ Filer/Non-Filer tracking for Pakistan tax compliance
- ✓ CNIC format display for national ID verification

- 🔄 Atomic transactions: multi-table updates succeed/fail together
- 💰 Automated financial calculations reduce manual errors
- 📋 Complete audit trail via ledger for every monetary movement
- 🌐 Localized features: PKR currency, CNIC, filer status

- 🔒 Prepared statements prevent SQL injection everywhere
- ✓ Business rule validation before state changes
- 📝 Transaction logging for regulatory compliance

 Module & Features **Key Capabilities** **Security & Access** **Print & Export Capabilities**

- ✓ Professional invoice layout with company branding
- ✓ Thermal & letterhead compatible receipt formatting
- ✓ Ledger print view with running balance
- ✓ Report print optimization hiding filters & actions
- ✓ CSV export for inventory & reports with applied filters
- ✓ Bike detail view with timeline for customer handover

-  Brand-consistent print layouts with logo & contact
-  Print media queries hide non-essential UI elements
-  One-click export for external accounting integration
-  Image inclusion in printed bike details

-  Print access controlled by page permissions
- ✓ Export data validation matches on-screen display
-  Export operation logging for data governance

 Search & Discovery

- ✓ Global inventory search: chassis, motor, model, color
- ✓ Customer search: name, phone, CNIC
- ✓ Advanced filtering on all list views: status, date range, category
- ✓ DataTables column search with regex support
- ✓ AJAX chassis uniqueness check during entry

-  Instant search results with client-side filtering
-  Precise filtering reduces data overload
-  AJAX validation prevents duplicate chassis entry
-  Mobile-optimized search inputs

-  Search scope limited by user role permissions
- ✓ Input sanitization prevents XSS in search results
-  Search query logging for usage analytics

 Performance & Reliability

- ✓ Optimized database queries with indexes on foreign keys
- ✓ Client-side caching via localStorage for theme/sidebar state
- ✓ Lazy image loading for inventory lists
- ✓ Efficient pagination on all data tables (default 100 rows)
- ✓ Background session timeout management
- ✓ Transaction rollback on multi-step operation failure

-  Fast page loads with minimal server round-trips
-  Smooth mobile experience with responsive assets
-  Graceful degradation on slow connections
-  Efficient memory usage via query optimization

-  Secure session handling with regeneration on login
- ✓ Data consistency via database transactions
-  Error logging via error_log for production debugging

 System Highlights **Category** **Key Achievements** **Security**

Enterprise-grade protection: bcrypt password hashing, CSRF tokens on all forms, SQL injection prevention via prepared statements, IP-based rate limiting with auto-ban,

 Category	 Key Achievements
	CAPTCHA with dynamic math problems, session timeout management, file upload validation, role-based access control with granular permissions
 Accessibility	Fully responsive mobile-first design, RTL-ready CSS structure, keyboard-navigable forms, high-contrast dark/light themes, print-optimized layouts, intuitive iconography with text labels, modal forms reduce navigation complexity
 Business Intelligence	Real-time margin/profit calculations, interactive Chart.js visualizations, model-wise stock analytics, tax collection reporting, installment overdue tracking, customer/supplier ledger with running balances, daily/monthly financial summaries
 User Experience	Professional dark-theme aesthetic with accent colors, SweetAlert2 confirmations for critical actions, JustValidate form feedback, Select2 enhanced dropdowns, DataTables with export/pagination, Animate.css micro-interactions, toast notifications for instant feedback
 Data Integrity	Double-entry ledger accounting, FOREIGN KEY constraints for referential integrity, atomic transactions with rollback, status workflow enforcement, stock validation on sales, chassis uniqueness checks, audit-ready transaction history
 Reliability	Session management with idle/absolute timeouts, database backup/restore functionality, error logging for production debugging, graceful handling of partial failures, input sanitization at all layers, comprehensive permission validation

PK Pakistan-Specific Features

 Feature	 Localized Implementation
Currency	PKR (Rs.) formatting with 2 decimal places, configurable symbol
Tax Compliance	Filer/Non-Filer customer tracking for FBR reporting, flexible tax calculation base
National ID	CNIC field with format hint (XXXXX-XXXXXXX-X), monospace display for readability
Payment Methods	Cheque workflow with bank name, cheque number, date, status (pending/cleared/bounced)
Phone Format	Pakistan mobile format support (03XX-XXXXXXX) in customer/supplier contacts
Address Fields	Province/city structured addresses (Punjab, Dera Ghazi Khan, etc.)

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Building innovative business management solutions for Pakistan's automotive retail sector

 Technical Stack

Layer	Technology
Backend	PHP 7.4+ with MySQLi, Prepared Statements
Database	MySQL 5.7+ with InnoDB, utf8mb4 charset
Frontend	HTML5, CSS3 (CSS Variables), Vanilla JavaScript
Libraries	jQuery 3.6, SweetAlert2, Select2, JustValidate, Chart.js, DataTables, Animate.css
Security	password_hash(), CSRF tokens, htmlspecialchars(), session security flags
Deployment	Shared hosting compatible, no framework dependencies



This report reflects all features, modules, and functionalities implemented in the BNI Enterprises Bike Dealer Management System as of version 2.0.0. All features are production-ready with comprehensive input validation, error handling, and audit trails. 